

AI Mastery for Modern Professionals

Module 1: AI Foundations & Core Concepts



Modules at a Glance

4 sessions

What we'll cover in this module

Orientation — the AI landscape

AI terminology demystified

Types of machine learning

Data — the fuel of AI

NLP, LLMs & Generative AI

AI in the real world — industry use cases

Session-1 Objectives

By the end of this session, you will be able to...



Define

AI and clearly distinguish between AI, ML, Deep Learning, and Generative AI



Describe

the evolution of AI from the 1950s to today's Large Language Models



Identify

real-world AI applications in their own industry and daily tools



Articulate

why AI matters for business and dispel at least 2 common AI myths



Feel confident

and curious — ready to build on this foundation in Sessions 2–26

Let's Start With You

Share with the group:

1 Your name & role

2 One AI technology you use

The Reveal:

Every tool you listed today likely uses AI under the hood.

You are already an AI user.

Today we understand what's happening.

**In one word — what comes to mind
when you hear the word AI?**

The AI Family — Understanding the Hierarchy

These terms are often confused. Let's fix that.

Artificial Intelligence

Any machine that mimics human intelligence

Machine Learning

Systems that learn patterns from data

Deep Learning

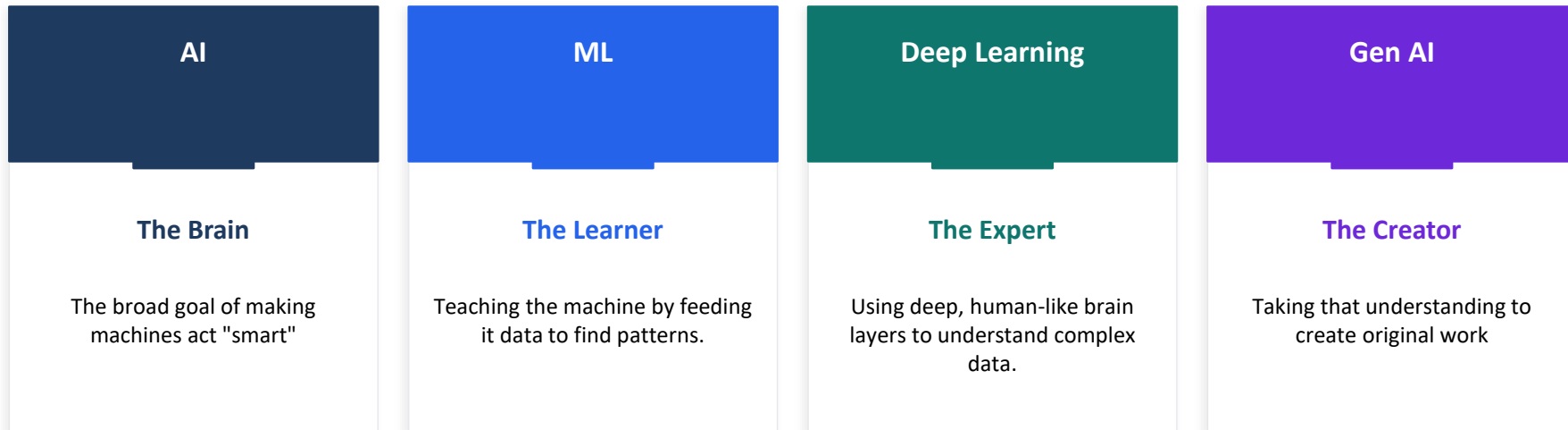
Neural networks with many layers

Generative AI

Creates new content — text, image, code

ChatGPT · Claude · Gemini · Copilot · DALL-E · Midjourney

A Plain-English Analogy



The Evolution of AI

From Turing's theory to ChatGPT — 70 years



The pace of change is not slowing — it is accelerating. This is why we are here today.

INTERACTIVE QUIZ-1

4 questions

Quiz —

Read each example carefully and identify which category it belongs to: AI , ML, DL, GenAI

A banking system detects suspicious credit card transactions by learning spending patterns over time.

ML

A self-driving car recognizes pedestrians, traffic signs, and road lanes using camera input.

DL

An AI tool generates realistic images from text prompts like “Create a futuristic city at sunset.”

Gen AI

A factory robot follows fixed programmed instructions to assemble products.

AI

AI Myth vs Fact

Let's address the elephant in the room

MYTH AI will replace all jobs in 5 years

FACT AI changes the **nature of roles** — judgment, creativity & relationships remain human

MYTH You need to code to use AI

FACT **No-code tools** are already powerful enough for most business use cases

MYTH AI is always accurate and truthful

FACT AI **'hallucinates'** — always verify outputs before using in decisions

MYTH AI understands and thinks like us

FACT AI **pattern-matches** on data — it has no understanding, feelings or common sense

MYTH AI is one single technology

FACT AI is an **umbrella** for ML, NLP, computer vision, reinforcement learning, and more

Why AI Matters — Right Now

The numbers, and what they mean for you

77%

of devices globally
use AI in some form

\$15.7T

AI's projected economic
contribution by 2030 (PwC)

50%

of companies have AI
in at least one function

Finance

Fraud detection, forecasting, reporting automation

HR

CV screening, engagement analytics, onboarding bots

Marketing

Personalisation, content generation, campaign analytics

Operations

Predictive maintenance, logistics optimisation, quality control

Sales

Lead scoring, pipeline forecasting, email personalisation

Legal

Contract review, compliance monitoring, risk summarisation

AI in Banking — Use Cases by Function & Technology

Operations & Systems · Database Administration · Automation & DevOps

Phase 1 · Session 1

Clover protects Knowledge Arm

	AI	ML	Deep Learning	Generative AI
Operations & Systems (OS)	 Intelligent system monitoring - Rule-based alert routing & auto-escalation - Capacity planning using IF-THEN logic - Automated failover decisions on threshold breach	 Predictive ops & anomaly detection - Server failure prediction before outage occurs - Network traffic anomaly detection in real time - CPU/memory trend forecasting for proactive scaling	 Deep pattern recognition in infra - Log sequence analysis across millions of events - Intrusion detection from raw network packets - Self-healing systems: detect + diagnose + remediate	 GenAI-assisted operations - Auto-generate incident reports from system logs - NL chatbot for ops team: 'Why is prod slow?' - Generate runbooks & resolution guides on demand
Database Admin (DBA)	 Rule-based data governance - Automated PII tagging using defined classification rules - Compliance checks: flag records missing mandatory fields - Scheduled integrity jobs with if-fail → alert logic	 Intelligent query & data management - Slow query prediction and auto-index recommendation - Data quality scoring: detect anomalies in financial records - Auto-classify documents (KYC, AML, contracts) from content	 Unstructured data intelligence - Extract fields from scanned cheques, forms & PDFs - Detect fraud patterns across billions of transactions - Read and index handwritten loan applications via OCR	 GenAI for DBA productivity - NL to SQL: 'Show top 10 customers by balance' - Auto-generate schema documentation from table definitions - Draft data migration scripts from change requests in plain English
Automation & DevOps	 Rule-based banking automation - Scheduled batch jobs: EOD reconciliation, interest calc - SWIFT message routing by format rules - Automated regulatory report submission on trigger dates	 Intelligent process automation - Real-time transaction fraud scoring at point of payment - Loan eligibility pre-screening from application data - AML risk scoring: flag suspicious patterns for review	 Deep learning automation - Automated cheque processing: read, validate, post - Voice biometrics for call centre authentication - Trade surveillance: detect market manipulation patterns	 GenAI-powered automation - Auto-draft regulatory filings from structured data - Generate customer communication for any account event - Copilot for DevOps: generate CI/CD scripts from intent

AI = Rule-based logic · ML = Learns from data · Deep Learning = Neural networks on large datasets · Generative AI = Creates new content / text / code

INTERACTIVE QUIZ-2

5 questions

Quiz — Question 1 of 5

we'll reveal the answer together

**"All AI is Machine Learning."
True / False ?**

FALSE

Answer: FALSE — Machine Learning is a subset of AI. AI also includes rule-based systems, expert systems, and other non-ML approaches. ML is one way to do AI — not the only way.

Quiz — Question 2 of 5

**"ChatGPT is an example of Generative AI."
True / False ?**

TRUE

Answer: TRUE — ChatGPT uses a Large Language Model (GPT-4) to generate new text responses. It is a flagship example of Generative AI — specifically text generation.

Quiz — Question 3 of 5

**"Deep Learning requires massive amounts of data."
True / False ?**

TRUE

Answer: TRUE — Deep Learning models have millions or billions of parameters that need significant data to train reliably. Smaller datasets are better suited to traditional ML or transfer learning.

Quiz — Question 4 of 5

**"Generative AI can only produce text."
True / False ?**

FALSE

Answer: FALSE — GenAI can produce text (ChatGPT), images (DALL-E, Midjourney), audio (ElevenLabs), video (Sora), and code (Copilot). The 'generative' refers to creating new content, not the format.

Quiz — Question 5 of 5

Vote on Mentimeter, then we'll reveal the answer together

**"You need to know coding to use AI tools effectively."
True / False ?**

FALSE

Answer: FALSE — This programme proves otherwise. No-code tools — ChatGPT, Copilot, Power Automate, Vertex AI AutoML — are powerful enough for most business use cases. Knowing how to prompt well matters more than coding.

3 Things to Take Away Today

1**AI is a family of technologies**

AI, ML, Deep Learning, and GenAI are related but distinct. You now know the difference.

2**You are already an AI user**

Gmail, Google Maps, Spotify, your phone keyboard — all powered by AI. The learning starts from familiarity.

3**AI is not magic — it's patterns**

AI finds patterns in data at massive scale. It doesn't think or understand. That distinction matters for how we use it.

Session 2: — AI Terminology Demystified

Thank You